VACANCY NOTICE – 2026-JRC.D.1-IPR-FGIV-003289

Science for policy researcher for soil nutrients, LUCAS soil monitoring and laboratory analysis

Type of contract	Member of the European Commission's contract staff, Function Group IV (article 3b of the Conditions of Employment of Other Servants)
Duration of contract	36 months (renewable up to maximum 6 years)
Area	Soil Health, Land Monitoring, Indicators development
Place of employment	Ispra (IT)
Indicative basic salary	4.449,31 - 6.444,59 (applicable as of 1st of July 2025) For more detailed information please consult: Working Conditions

WE ARE

The Joint Research Centre (JRC) provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society.

The current vacancy is in the **Directorate for Sustainable Resources**. Its science, positioned at the crossroads of the environment, economy and society, coupled with scientific networks across the EU and beyond, informs policies to ensure that present and future generations enjoy a sustainable planet.

The vacancy is in **Land and Climate Unit**. Its cross-disciplinary science provides independent evidence in support of EU policies and international conventions on land and climate. It assesses land resources and the terrestrial carbon cycle, while analysing options for the sustainable management of soil and forest ecosystems in view of human and increasingly climate-related pressures, to secure our natural resources and carbon sinks. The post is located in Ispra.

The successful candidate will join the EU Soil Observatory (EUSO) project.

More details can be found at: https://esdac.jrc.ec.europa.eu/euso.

The activities will focus on soil monitoring and development of soil health indicators relevant to soil nutrients to assess progress on soil health in the EU. The successful candidate will contribute to the development of an integrated EU Soil Monitoring system and support the implementation of the Soil Monitoring Law. The successful candidate will conduct scientific research on advancing EU soil monitoring and indicator development by combining the LUCAS Soil Module of the European Commission with national monitoring systems in Member States, future data flows from the Soil Monitoring Law implementation, and new data from Mission Soil projects. This research will not only deepen our scientific understanding of soil health but also underpin EU policies and Commission contributions to assess the sustainable competitiveness of the land sector and inform soil, agricultural, climate and biodiversity policies and debates.

We offer high-quality working conditions in a dynamic environment and in a multidisciplinary and multicultural team. The JRC encourages a collaborative workplace culture and interactions with policy makers, academics and other leading stakeholders active in the field are foreseen. The job includes participation in (international) conferences and (hybrid) workshops, as well as occasional travel.

Please see also Working at the Commission – conditions and environment (europa.eu)

WE PROPOSE

The jobholder will contribute to the development of an integrated EU Soil Monitoring system and advance in soil nutrients indicator development in the context of Soil Monitoring Law and the HORIZON Mission "A Soil Deal for Europe". Research will focus on novel approaches to develop soil nutrients indicators and to combine them in dashboards and tools for policy support, such as the EU Soil Strategy for 2030, the Soil Monitoring Law, and the Nitrates Directive. In addition, the jobholder will advance in reference methodologies, laboratory standards and results from LUCAS soil, and development of transfer functions for a harmonised soil monitoring framework in the European Union.

The jobholder will mainly concentrate on:

1) Development of models, indicators, dataset and tools to assess the nutrient (NPK) content in EU topsoils based on LUCAS topsoil and other sources (e.g. Soil Mission projects, Soil Monitoring Law data flows, Statistical data, etc.).

2) Providing scientific and technical support to Member States and EU Commission services (DG ENV, DG AGRI, EUROSTAT, DG GROW) for soil related EU policies such as the Soil Monitoring Law, the EU Soil Strategy 2030, the Nitrates Directive, Regulation on statistics of agricultural input and output (SAIO), Common Agricultural Policy (CAP) and the EU Fertilizer Action Plan.

3) The jobholder should perform a quality assessment of LUCAS soil data, and develop transfer functions addressing issues as different sampling depths, analytical methods and protocols contributing to the development of EU Soil harmonised monitoring system as described in the third building block of the Soil Mission Implementation Plan. Provide technical support for implementation of technical laboratory specifications, implementation and development of European and International standards and reference methodologies in laboratory analysis of physical and chemical properties for the support of the Soil Monitoring Law implementation and for the LUCAS soil survey.

The jobholder will engage with DG ENV, DG AGRI, DG GROW, EUROSTAT Member States, HORIZON projects and EUSO stakeholders to drive forward above-mentioned developments. The jobholder will be responsible for synthesizing research findings into high-quality scientific articles, dashboards, conference or project meeting presentations, policy briefs, and technical reports.

WE LOOK FOR

We are looking for applicants with a background in soil science, natural sciences, environmental sciences, agricultural sciences who have experience in indicators development and geospatial data processing. The successful candidate should be passionate about

advancing soil health assessments and soil monitoring, including the spatial and temporal data analysis of maps.

The successful candidate will have:

- A PhD in soil sciences, Earth sciences, environmental sciences, biology, ecology, agricultural sciences or related issues. Alternatively, a university degree in those fields and at least 5-years proven job-related experience in a field relevant to the position.
- Previous research or professional experience in the field of soil indicators development, soil nutrients (Nitrogen, Phosphorus, Potassium) assessments and soil monitoring. The experience should be proven and cover at least 1 year of activity.
- Modelling experience in soil nutrients dynamics in agroecosystems with particular specialization in machine learning (e.g. Digital Soil Mapping) or Bio-geochemical model (e.g. Century, DayCent or equivalent).
- Extensive knowledge/experience in geospatial data processing (e.g. ArcGIS or similar open source) / database skills and programming/scripting capacity (e.g. R, Python)
- Experience in Machine learning, deep-learning, artificial intelligence and advanced statistical inference for development of pedo-transfer functions
- Excellent knowledge of oral and written English (C1).
- a solid record of research activities, including relevant publications in international peer-reviewed journals.

Desirable assets are:

- Experience and knowledge on technical laboratory specifications, ISO standards and reference methodologies in laboratory analysis of physical and chemical properties
- Experience in distributed or cloud computing;
- Experience in developing Dashboards and composite indicators

The successful candidate should be able to communicate effectively, have the ability to translate technical results into actionable insights for policymakers and stakeholders, and be able to synthesize research findings into high-quality scientific articles.

If you are innovative, detail-oriented, and eager to contribute to impactful soil and natural land research in a larger policy environment, we encourage you to apply.

HOW TO APPLY

If you are **already on a valid CAST FG IV** reserve **list**, or you **have already applied to one of the calls below**, you can directly submit your application at JRC Recruitment Portal.

If not, before applying to this position, **you must register** for one of the two following databases:

- the Call for Expressions of Interest | EU Careers (europa.eu) (CAST Permanent FG IV), which is used by a wide range of organisations (institutions, bodies, offices and agencies of the European Union), or
- the call for researchers (JRC Call COM/1/2015/GFIV – Research), which is mainly used by the JRC.

Note that each of the calls above has **different minimum eligibility requirements and different selection tests**.

The JRC cultivates a workplace based on respect for other people and the environment, and embraces non-discriminatory practices and equality of opportunity. In case of equal merit, preference will be given to the gender in minority.